Every matching is obtained from its core by successive insertions of short chords

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Editorial note. The counted statement is the author's labelled lemma 'lem:matching is generated by inserts to its core' (source lines 1190-1196); the proof environment begins on the next line after \end{lemma} apart from one blank line, so the pairing is by adjacency.

Editorial note. The retained context is the author's own sentence position immediately preceding the statement inside the section developing the insertion calculus.

Editorial note. The retained preamble is the editorially prepared standalone preamble retained for this source; packages needed only by other parts of the article are left out because no retained line uses them.

Editorial note. No retained line refers to a label, citation or figure outside the retained span, so the package declares no external reference.

Editorial scope. One original result of the pinned publisher article, counted once: generation of a matching from its core by insertions of short chords (author label lem:matching is generated by inserts to its core).

Allowed standard prerequisites (editorial).

- Matchings and unstable chords. M_N is the set of perfect matchings of [2N] written as a set of chords (i,j) with i < j; a chord is stable if it is short in the sense fixed in the section, and k counts the unstable chords.. The statement is an induction on the number k of unstable chords.
- Insertion of a short chord. For a matching m in M_{N-2} and an index i, insrt_i(m) is the matching of M_N obtained by adding the short chord (i,i+1) and shifting the labels of m above i by 2.. It is the operation used both in the statement and in the induction step at source line 1201.
- Core of a matching. core(m) is obtained by repeatedly deleting short chords and is a matching without unstable chords.. The induction step uses that deleting one short chord does not change the core, core(m) = core(m_1).

Lemma 0.1. Let $m \in \mathcal{M}_N$ be a matching with k unstable chords. Then there exist indices $i_1, \ldots, i_k$ such that

$$m = \mathrm{insert}_{i_1} \cdots \mathrm{insert}_{i_k}(\mathrm{core}(m)).$$

Proof. We prove the statement by induction on k. If $k = 0$ then $\mathrm{core}(m) = m$ and the statement is obvious.

Assume that a matching $m \in \mathcal{M}_N$ has $k > 0$ unstable chords. Therefore, m has at least one short chord, denoted $(i_1, i_1+1) \in m$. Thus, there exists a matching $m_1 \in \mathcal{M}_{N-2}$ such that $m = \mathrm{insert}_{i_1}(m_1)$. Notice that $\mathrm{core}(m) = \mathrm{core}(m_1)$, so m_1 has $k-1$ unstable chords. Thus, by the induction hypothesis, there exist indices $i_2, \ldots, i_k$ such that $m_1 = \mathrm{insert}_{i_2} \cdots \mathrm{insert}_{i_k}(\mathrm{core}(m))$ and so $m = \mathrm{insert}_{i_1} \cdots \mathrm{insert}_{i_k}(\mathrm{core}(m))$. □